

QKV

Scaled Dot-Product Attention A Worked Example

A one-chapter textbook built from one section of one paper

Before You Read

This is a one-chapter textbook built from a single source: Vaswani et al. (2017), "Attention Is All You Need" (arXiv:1706.03762), limited to the abstract and Section 3.2.1 (including that section's footnote 4).

Sentences that make claims about the paper carry an evidence grade. `[confirmed]` means the paper states it directly. `[general]` means an established fact of linear algebra or probability. `[inferred]` means a reading or example that follows from confirmed or general statements but that the paper does not state itself. `[not found]` means the claim was not found in the part of the paper used here (the abstract and Section 3.2.1); other sections of the paper may still contain it.

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PART 01

Part 1

Understanding Scaled Dot-Product Attention

01

Scaled Dot-Product Attention

Learning objective.

Understand what Q, K, and V each stand for well enough to explain, one step at a time, what the formula $\text{softmax}(QK^T/\sqrt{d_k})V$ computes.

Reading the source. “We call our particular attention "Scaled Dot-Product Attention" (Figure 2). The input consists of queries and keys of dimension d_k , and values of dimension d_v . We compute the dot products of the query with all keys, divide each by $\sqrt{d_k}$, and apply a softmax function to obtain the weights on the values.”¹

[confirmed] The paper writes this computation as $\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$.¹

1.1 Queries, keys, and values

Concept

queries, keys, and values (Q, K, V) Q, K, and V are all matrices, each a stack of vectors [confirmed]. A query (Q) can be read as a vector for "what this position is looking for", a key^s (K) as one for "what each position has to offer", and a value^s (V) as one for "the content that position actually hands over" [inferred]. Here a "position" is where one word of the sentence sits. This book's reading of why there are three roles: to adjust the matching rule (Q against K) and the content that gets mixed in (V) separately during training, they have to be kept apart [inferred]. Example: in the sentence "She sat on the bank of the river", the query for "bank" asks for clues that settle which meaning of "bank" is meant; if it matches the key for "river" best, the value for "river" makes up a large share of what "bank" receives [inferred].

Quick check. If the query and the value were the same vector ($Q = V$), what would the design lose? Answer: "what to look for" and "what to hand over" could no longer be adjusted separately.

1.2 The dot product

Concept

the dot product^s (QK^T) The dot product of two vectors multiplies their matching components and adds up the results [general]. The result is larger the more the two vectors point the same way and the longer the vectors are [general]. QK^T computes the dot product of every query with every key in one step, producing a table (a matrix) of numbers [confirmed]. Each entry of that table can be read as "how well one query matches one key" [inferred].

Quick check. With 5 queries and 7 keys, how many rows and columns does QK^T have? Answer: 5 rows and 7 columns, one dot product per query-key pair.

1.3 Scaling

Concept

scaling (dividing by $\sqrt{d_k}$) Dot products tend to grow as the vector dimension d_k grows [general]. The paper's footnote 4 gives the reason: if the components of q and k are independent random values with mean 0 and variance 1 (variance is how spread out the values are), their dot product has variance d_k [confirmed].² The next sentence gives the reason for dividing: the authors "suspect" that large dot products push the softmax into regions where it has extremely small gradients (the gradient is how much the output changes when the input changes a little) [confirmed]. When the gradient is that small, training gets only a weak signal for correcting the values [general]. The section does not directly test the suspicion with an experiment or a proof [not found].

Quick check. If d_k grows from 4 to 400, the divisor $\sqrt{d_k}$ grows from 2 to what?
Answer: 20.

1.4 Softmax

Concept

softmax Softmax takes a list of numbers and turns it into a list of positive numbers that add up to 1, with a larger input getting a larger share [general]. In `Attention(Q, K, V)`, softmax turns each query's row of match scores into weights that say how much attention each key gets [confirmed]. Multiplying these weights by the values (V) and adding them up gives the output for that query's position [confirmed] (the paper's Equation 1).

Exercise 1. A reader who has never worked out what dividing by $\sqrt{d_k}$ does to the size of the softmax inputs can only answer "why is this division needed?" by reciting the formula.

1. Take two vectors with $d_k = 2$ whose components are all 1. Compute their dot product, then divide it by $\sqrt{d_k}$.

2. Do the same for two vectors with $d_k = 200$ whose components are all 1.
 3. When d_k goes from 2 to 200, by what factor does the dot product grow, and by what factor does the divided value grow?
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Review 1.

1. Which of Q, K, and V stands for "what to look for"? (a) Q (b) K (c) V
2. Each entry of QK^T is the () of two vectors.
3. After softmax, what do the resulting values always add up to?

Answers: Appendix A.

Glossary

query (Q) First seen p.5 — A vector for what the current position is looking for.

key (K) First seen p.6 — A vector for what each position has to offer.

value (V) First seen p.6 — A vector for the content a position actually hands over.

dot product First seen p.6 — The sum of the products of two vectors' matching components; it grows the more the two vectors point the same way and the longer they are.

d_k First seen p.5 — The dimension (number of components) of the query and key vectors. The dimension of the value vectors is written d_v.

softmax First seen p.5 — A function that turns a list of numbers into a list of positive numbers (shares) that add up to 1.

APPENDIX

Appendix

Reference material

Appendix A. Answers

Chapter 1, Exercise 1.

1. $1 \times 1 + 1 \times 1 = 2$. Divided by $\sqrt{2}$, about 1.41.
2. 1×1 added 200 times = 200. Divided by $\sqrt{200}$, about 14.1.
3. The dot product grows $100\times$ ($2 \rightarrow 200$) and the divided value grows $10\times$ ($1.41 \rightarrow 14.1$, which is $\sqrt{100}$). The division does not stop the growth, but it slows it from $100\times$ to $10\times$.

Chapter 1, Review. 1) (a) Q 2) dot product 3) 1

Appendix B. Footnotes

- 1 Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin, "Attention Is All You Need," arXiv:1706.03762 (2017), §3.2.1. [↗](#)
- 2 Same paper, §3.2.1, footnote 4: "To illustrate why the dot products get large, assume that the components of q and k are independent random variables with mean 0 and variance 1." [↗](#)